

Major Production Requirements: Enterprise Web App Upgrades

This document outlines the major architectural requirements and advanced features needed to transition HireSense from a single-user ATS platform into a production-grade, enterprise-ready B2B SaaS recruiting suite.

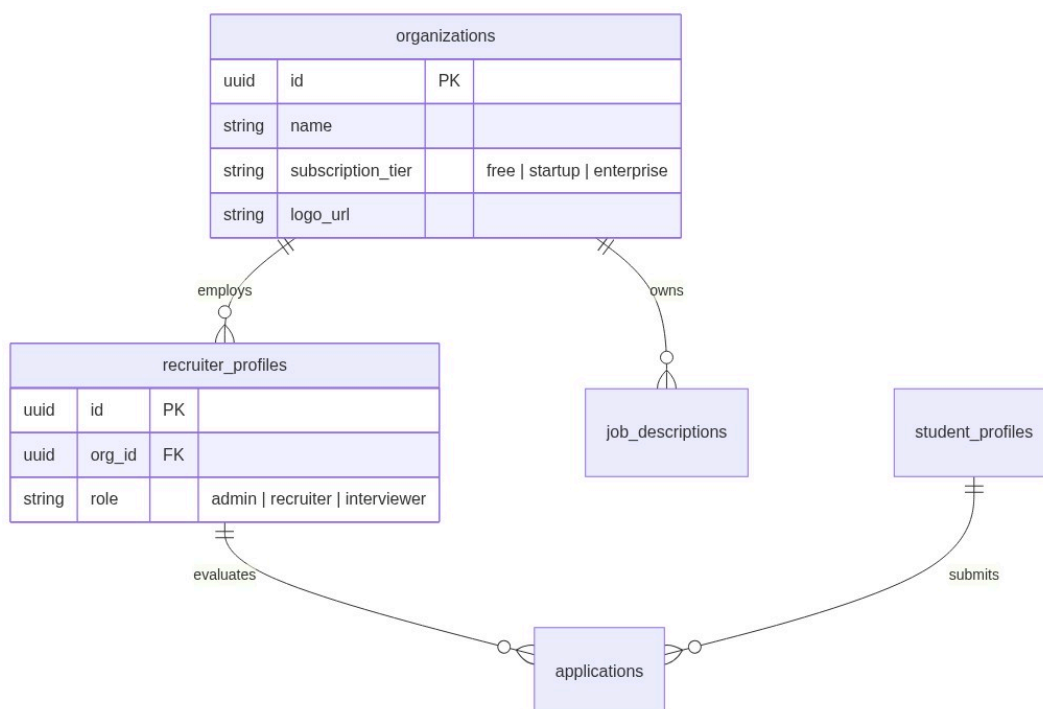
1. Multi-Tenant Organization Structure

Current Status

Data is strictly isolated on a per-user basis (`user_id`). Recruiters cannot collaborate or share candidate pools.

The Requirement

Introduce an Organization/Tenant layer. Recruiters work in shared workspaces corresponding to their company, while students remain in a global candidate pool.

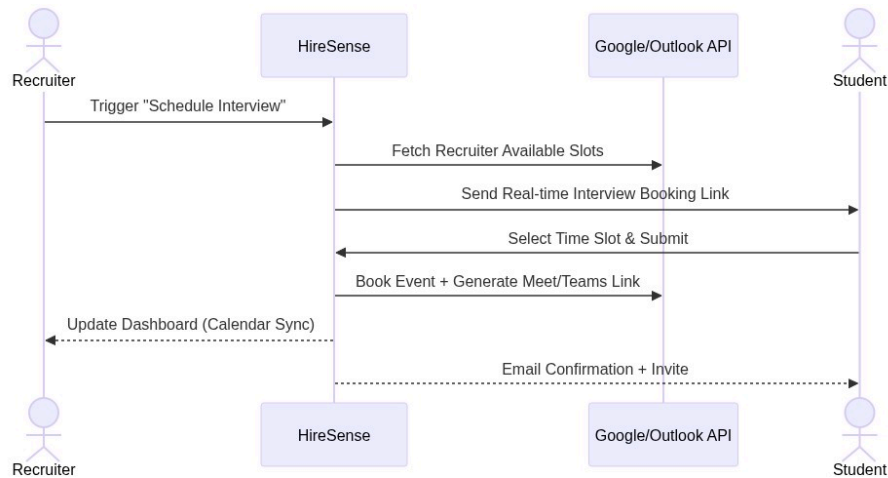


- **Recruiter Collaboration:** Allows multiple recruiters from the same company to view, score, and comment on the same candidate pipeline.
- **Shared Talent Pool:** Enables organizations to keep a shared database of historic applicants, searchable via semantic vector queries.

2. Interview Scheduling & Calendar Integration

The Requirement

Recruiters need to schedule interviews without leaving the platform. This requires integration with third-party calendar providers (Google Workspace and Microsoft Outlook).



- **Interactive Booking:** Students receive a personalized booking link showing the recruiter's real-time availability (similar to Calendly).
- **Automatic Video Links:** Automatically generates Google Meet or Microsoft Teams links upon selection.
- **Pipeline Transition:** Automatically moves candidate status in the pipeline to interviewing upon successful booking.

3. Enterprise ATS Syncing (Greenhouse, Workday, Lever)

The Requirement

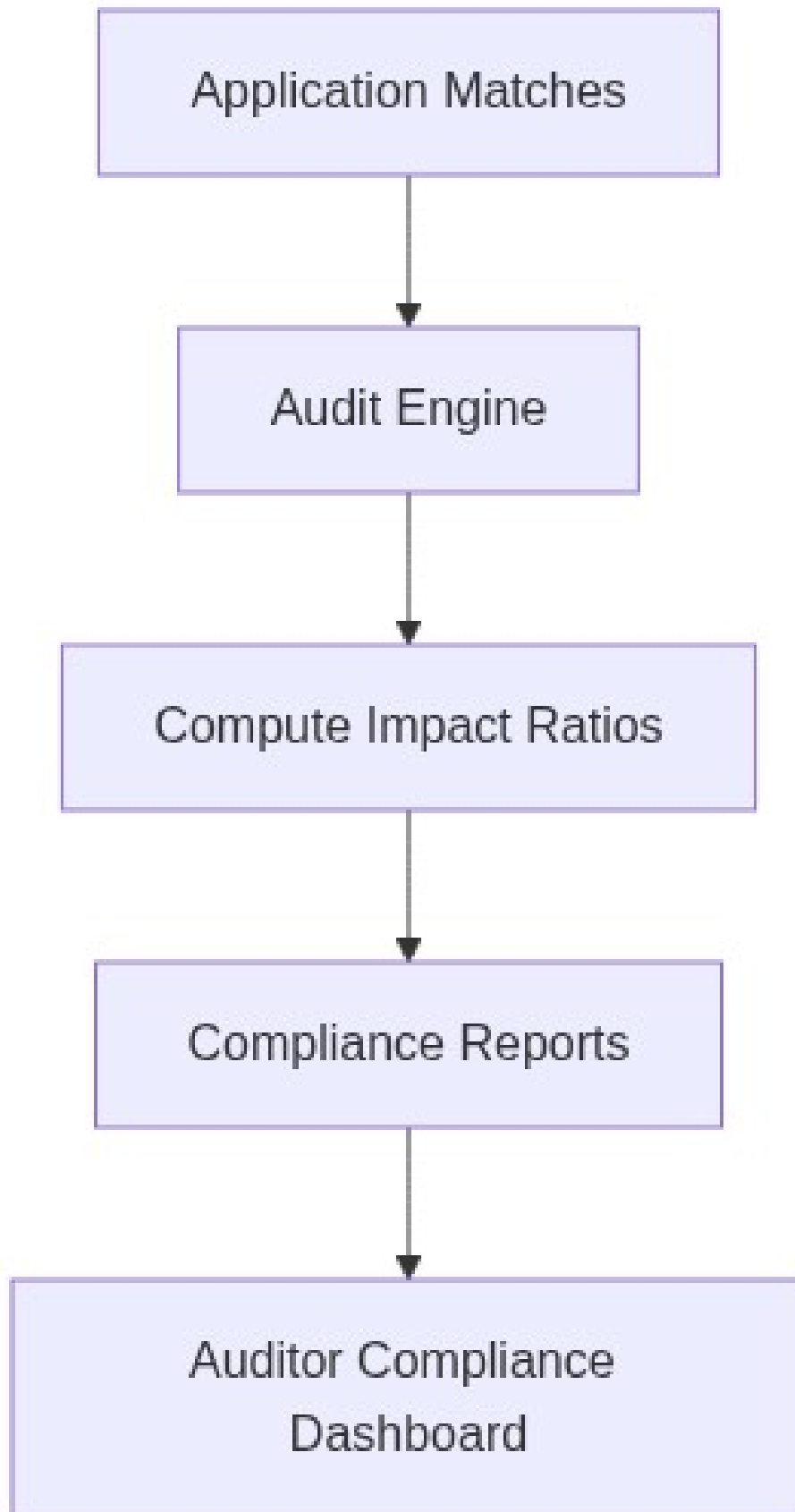
Mid-market and enterprise companies rarely use a standalone portal; they require integrations with their existing "Source of Truth" Applicant Tracking Systems.

- **Two-Way Synchronization:**
- **Pull Jobs:** Periodically import job descriptions published on Greenhouse/Workday into the HireSense portal.
- **Push Candidates:** When a student applies via the Student Portal and is parsed, push their profile, resume PDF, and AI matching score directly into the company's enterprise ATS pipeline.
- **Webhooks Integration:** Listen to hiring stage changes inside Greenhouse to trigger real-time updates inside HireSense.

4. AI Bias Compliance & Auditing Center

The Requirement

Global jurisdictions are introducing strict legislation on Automated Employment Decision Tools (AEDT) (e.g., New York City Local Law 144 and the EU AI Act). Enterprise platforms must provide auditable proof of non-bias.



- AEDT Compliance Reports: Automatically calculate and monitor "Impact Ratios" across demographic markers (gender, ethnicity, age) to prove the AI model scoring is statistically fair.
- Explanation System: Provide clear, deterministic rationale logs explaining how the matching algorithm

computed specific sub-scores to protect recruiters against compliance audits.

- Redaction Pipeline: Optional strict PII removal that strips names, addresses, graduation years, and gender pronouns from candidate profiles before passing data to LLM engines.
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5. Role-Based Access Control (RBAC)

The Requirement

Recruiters, department heads, and interviewers require different levels of visibility. A flat security model exposes sensitive data and settings.

- RBAC Roles:
 - Super Admin: Manages organization details, billing, API keys, and team member seats.
 - Recruiter: Creates job listings, reviews matches, moves candidates in pipelines, and schedules interviews.
 - Interviewer / Hiring Manager: Accesses candidate profiles assigned specifically to them, records interview feedback notes, and inputs scorecard ratings (cannot see other candidate pools).
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6. Observability & AI Telemetry

The Requirement

To operate reliably in production, engineering teams need visibility into network latencies, rate limits, model failures, and system exceptions.

- NVIDIA NIM Telemetry: Logs race success rates, model response latencies, and fallback frequencies to keep track of API costs and LLM response speed.
- System Telemetry:
- OpenTelemetry/Sentry: Traces backend routes and identifies network bottlenecks.
- Prometheus + Grafana: Monitors system CPU/Memory loads, storage usage, and DB replication connection pooling.